

CREDIT PORTFOLIO SECTOR CONCENTRATION AND ITS IMPLICATIONS FOR CAPITAL REQUIREMENTS

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This article assesses whether the sector concentration of the portfolio of loans to resident and non-resident legal entities according to information from the Central Credit Register (CCR) indicates a need for additional capital requirements under Pillar 2. A simplified version of the factor model is used for this purpose. The Herfindahl-Hirschman Index (HHI) is employed to indicate the degree of sector concentration and its long-term tendencies. The results of the factor model, which takes account of an institution's risk profile, signal a possible need for additional economic capital for portfolios featuring higher sector concentration and higher risk. In such cases, supervisors should evaluate whether sector concentration risk is correctly measured, assessed and incorporated into capital requirements in the bank's internal capital adequacy assessment process (ICAAP).

1. INTRODUCTION

Concentration of credit risk in asset portfolios is one of the potential causes of large losses in credit institutions documented in the literature (BCBS, 2004) and one of the specific risks subject to supervisory review under Pillar 2 (CEBS, 2010). The reason for this is that concentration risk is not fully reflected in the Pillar 1 minimum capital requirements, as credit portfolios do not satisfy the condition of perfect diversification as assumed by the IRB approaches to the measurement of credit risk (BCBS, 2006).

Concentration risk arises both at the level of individual large exposures (large exposure risk) and from excessive exposure to a single sector or a number of significantly correlated sectors (sector concentration). Large exposure risk is addressed in detail in the EU's Capital Requirements Regulation (CRR), which contains criteria and techniques for assessing it (Articles 387–403). Sector concentration, however, is not subject to detailed regulation.

This article therefore presents a possible methodological approach to this issue based on a simplified version of the factor model (see Düllmann and Masschelein, 2007) and applies that approach to evaluate whether the sector concentration of banks implies a need for additional capital requirements under Pillar 2. It also uses the Herfindahl-Hirschman Index as an alternative indicator of sector concentration risk and its long-term tendencies in order to identify possible new areas of risk accumulation at the macro-level of the banking sector as a whole.

2. REGULATORY FRAMEWORK FOR CREDIT CONCENTRATION RISK

Article 81 of Directive 2013/36/EU of the European Parliament and of the Council of 26 June 2013 on access to the activity of credit institutions and the prudential supervision of credit institutions and investment firms (CRD) requires competent authorities to ensure that the concentration risk arising from exposures to each counterparty, including central counterparties, groups of connected counterparties and counterparties in the same economic sector, in the same geographic region or from the same activity or commodity, is addressed and controlled, including by means of written policies and procedures.

Requirements for the evaluation of concentration risk by supervisory activities are also laid down in Articles 144–151 of the *Guidelines on common procedures and methodologies for the supervisory review and evaluation process* (SREP; see EBA, 2014). These guidelines require competent authorities to assess the nature of credit risk by considering at least: (a) credit concentration risk, (b) counterparty credit risk and settlement risk, (c) country risk, (d) credit risk from securitisations, (e) FX lending risk, and (f) specialised lending risk.

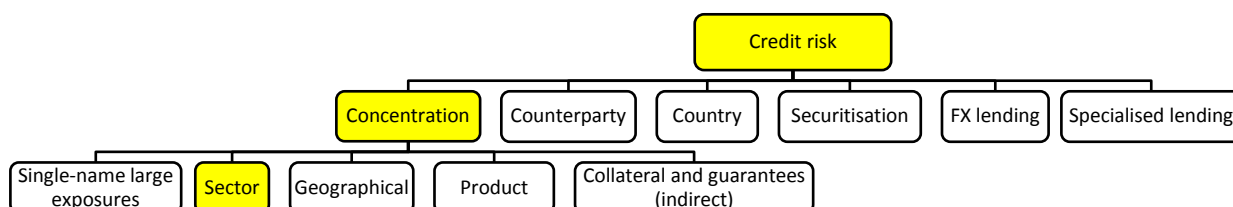
When evaluating concentration risk, competent authorities should form a view on the degree of risk that the institution will incur significant credit losses stemming from a concentration of exposures to a small group of borrowers, to a set of borrowers with similar default behaviour or to highly correlated financial assets.

Competent authorities should conduct this assessment considering different categories of credit concentration risk, including: (a) single-name concentrations (including a client

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FIGURE 1

CREDIT RISK BREAKDOWN



Source: Authors

or group of connected clients as defined for large exposures – see Articles 387–403 of CRR 575/2013/EU), (b) sector concentrations, (c) geographical concentrations, (d) product concentration, and (e) collateral and guarantees concentration. Figure 1 shows the position of sector concentration risk within the overall framework of credit risk.

Competent authorities can use various measures and indicators to assess the level of concentration. The most common is the Herfindahl-Hirschman Index (HHI), which can be incorporated into more or less sophisticated methodologies to estimate the additional credit risk impact.

An Advanced Working Group (AWG) on Sectoral Risk has been established at the EBA to unify the approach to measuring sector concentrations at EU level. At its meeting at the end of January 2015, the AWG presented a proposal for a single approach to measuring sector concentrations. The final document will be submitted in April 2015. The solution involves the use of the HHI (section 4 of this article) and a model-based approach to determining capital requirements associated with sector concentrations (section 5).

3. SECTOR CREDIT CONCENTRATIONS ACCORDING TO THE CCR

Concentrations can be evaluated from either a systemic or granular perspective. From the financial stability/systemic perspective it is vital to focus on the risks to groups of banks arising from common business concentrations. From the internal risk management and supervisory perspective (the individual micro-perspective) the most important thing is to concentrate on risks at the individual institution level.

In this article we analyse sector concentration at both the systemic and granular level. The effect of sector

concentration and its structure on economic capital requirements is documented using several portfolios. The benchmark portfolio reflects the sector concentration of the aggregate exposure to resident and non-resident legal entities, which report to the Central Credit Register (CCR). The model portfolios simulate characteristics of sector concentration in the banking sector.

In the interests of clear presentation of the analytical and modelling framework, we opted for sectoral breakdown at the level of the main industry sectors of the national economy. Industries with exposures of less than CZK 30 billion were pooled into the “Other” category. The part of the CCR credit portfolio for which no sector identification is available consists mainly of non-residents.

TABLE 1

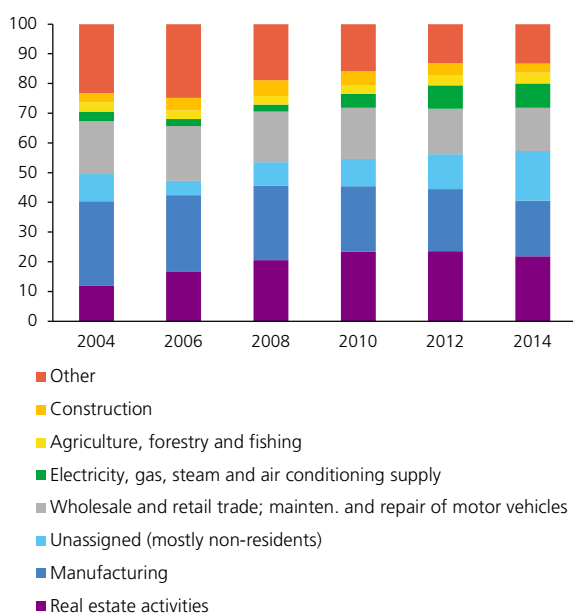
SECTOR CONCENTRATIONS OF THE BENCHMARK PORTFOLIO (CZK billions)

Sector	2004	2006	2008	2010	2012	2014
Real estate activities	77	131	192	210	233	254
Manufacturing	183	201	236	197	205	218
Of which: Manufacture of food products	25	24	29	23	23	19
Manufacture of metals and metal products	22	29	41	32	33	36
Manufacture of motor vehicles and other transport equipment	14	16	30	23	30	36
Manufacture of machinery and equipment	19	23	26	20	20	24
Other	103	109	110	99	99	103
Unassigned (mostly non-residents)	59	38	74	83	116	194
Wholesale and retail trade; maintenance and repair of motor vehicles	115	144	159	155	151	169
Electricity, gas, steam and air conditioning supply	21	19	23	43	77	96
Agriculture, forestry and fishing	20	22	27	26	35	42
Construction	20	33	49	42	39	36
Other	150	194	177	142	130	154
Total	645	783	937	898	985	1163

Source: CNB

CHART 1
SHARES OF SECTORS IN TOTAL CREDIT EXPOSURE

(shares in %)



Source: CNB

This segment is categorised as “Unassigned”. The sector concentrations of the benchmark portfolio in 2004–2014 are presented in Table 1 and the shares of industries in the total credit exposure are shown in Chart 1. The data on the internal structure of manufacturing, the industry with the second-largest total exposure, reveal that no subsector is dominant (only four subsectors have shares exceeding 10% of the total exposure) and the dynamics of the exposures are little changed since 2008.

The dominant industries from the exposure structure perspective are “Real estate activities”, “Manufacturing” and “Wholesale and retail trade; maintenance and repair of motor vehicles”. The “Unassigned” category is dominated by loans to non-residents, which saw sharp growth between 2012 and 2014, mainly because of an expansion in the foreign activities of some banking groups. The most important change is that in the dominant industry from “Manufacturing” in 2004 (28% of the total exposure) to “Real estate activities” in 2014 with a share of 22% (versus 19% for “Manufacturing”). The changes signal a need for increased attention both in supervisory practice and in the design of macroprudential policy. Assessed in the context of change in the industries’ contributions to production (Table 2), the relative shares of the main economic sectors have not altered significantly over time. This clearly reveals the potential risks.

TABLE 2
SHARES OF SECTORS IN PRODUCTION

(in %)

Sector	2004	2008	2013
Manufacturing	33	35	36
Wholesale and retail trade; maintenance and repair of motor vehicles	8	8	9
Construction	9	10	8
Real estate activities	6	6	7
Transport	6	6	6
Electricity, gas, steam and air conditioning supply	6	4	4
Agriculture, forestry and fishing	2	2	2
Other	30	29	28
Total	100	100	100

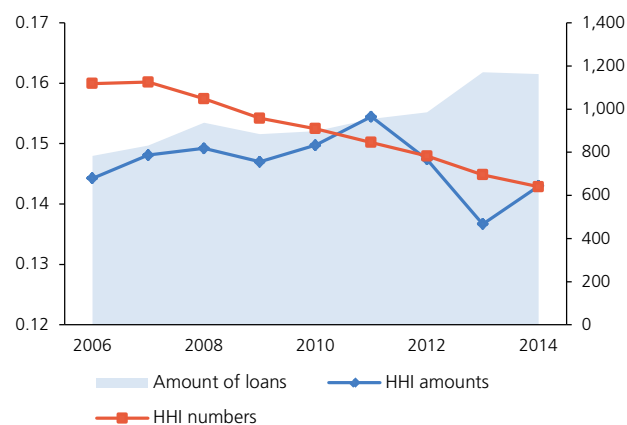
Source: CZSO

4. SECTOR CONCENTRATION AS MEASURED BY THE HHI

Use of the HHI is one of the recommended methods for evaluating sector concentration. This technique is a simple, non-model-based approach to determining undiversified idiosyncratic risk. The HHI is defined as the sum of the squared relative shares of borrowers’ sectoral exposures in the total size of the portfolio. Well diversified portfolios containing a large number of small firms have HHIs close to zero, while highly concentrated portfolios have much higher HHIs. In the extreme case of a single borrower the HHI equals one. For the purposes of this article we determine the HHI at the total sector exposure level. The results reveal the significance of individual portfolios from the concentration risk perspective. However, they do not take into account borrowers’ different probabilities of default,

CHART 2
SIZE OF EXPOSURES AND THE HHI OF THE BENCHMARK PORTFOLIO

(index on left-hand scale; exposures in CZK billions on right-hand scale)

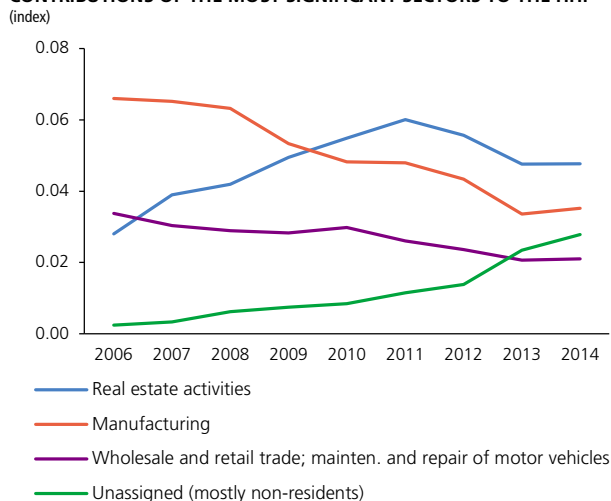


Source: CNB, authors’ calculations

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CHART 3

CONTRIBUTIONS OF THE MOST SIGNIFICANT SECTORS TO THE HHI



Source: CNB, authors' calculations

collateral and inter-sector credit risk correlations and hence cannot provide information on the economic capital needed to cover risks.

At the system level, the HHI of the benchmark portfolio is 0.14 (see Chart 2, HHI amounts), i.e. a generally quite low concentration. The index varies over time within a range of 0.135 in 2013 and 0.155 in 2011, when it peaked in value. The sizeable decline in 2013 was caused by the incorporation of new exposures into the CCR linked with an expansion in foreign activities by some banking groups.

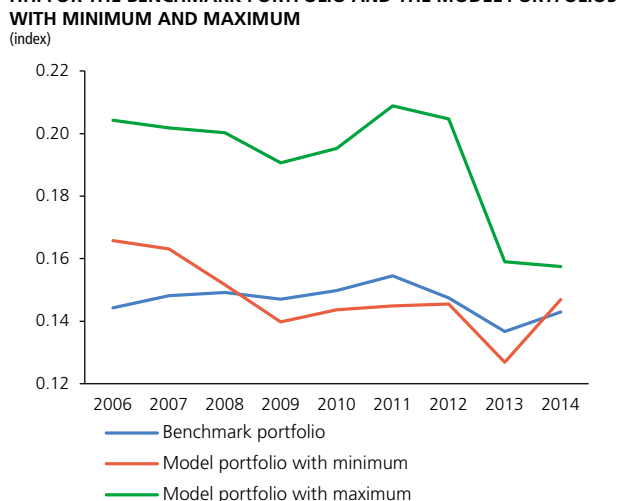
From the international perspective, the HHI takes similar values as in Germany (0.14 for regional banks and 0.17 for nationwide banks). However, the exposure share of "Real estate activities" differs, being around 15%¹ in Germany, i.e. 8 percentage points lower than in the Czech Republic (at the end of 2011).

The constant decline in the HHI as measured by the number of loans² in the period under review is due to higher sector diversification of new loans (regardless of amount).

The internal structure of the HHI from the sectoral perspective is shown in Chart 3, which depicts the change in the shares of industries in the total exposure. In 2010,

CHART 4

HHI FOR THE BENCHMARK PORTFOLIO AND THE MODEL PORTFOLIOS



Source: CNB, authors' calculations

"Real estate activities" became the sector with the largest exposure amid a steady decline in the weights of traditional industries. The significant growth in non-residents' exposures implies a need to pay increased attention to the risks associated with these exposures.

At a granular level, the HHI for the five largest banks fluctuates in a wide range of 0.13–0.21 (see Chart 4), signalling different approaches to the management of concentration risk and, for banks with higher HHIs, a need for more detailed analysis of all aspects of concentration risk. A positive tendency towards a decline in this measure of sector concentration and a narrowing of the range of the HHI can generally be observed since 2011.

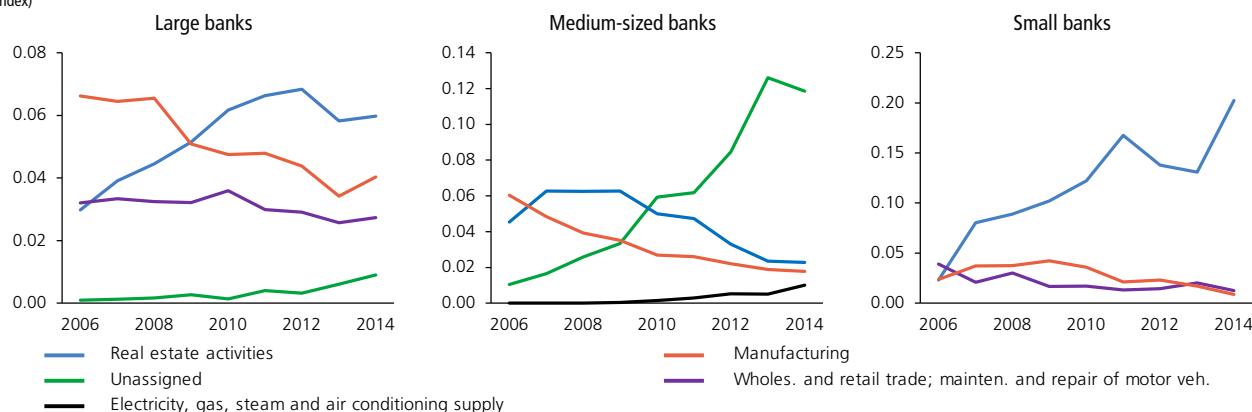
The change in the key industry in terms of concentration from "Manufacturing" to "Real estate activities" is observed for all bank segments (large banks, medium-sized banks and small banks – see Chart 5). However, after peaking in 2011, the HHI for "Real estate activities" starts to decline. The only exception is the small bank segment, which saw a sharp rise in 2014, possibly indicating higher risk. Likewise, there was a significant increase in exposure to non-residents ("Unassigned") in the medium-sized bank segment, which, given its dominant position, is growing in significance. In this case, though, one needs to be more cautious in assessing the situation, because the non-resident sector may be relatively diversified from the sectoral and geographical perspective. The current growth in geopolitical risks of a regional character highlights the need for deeper analysis to determine the degree of risk. In the medium-sized bank segment, the "Electricity supply" industry has

1 Jahn et al. (2013), p. 7, industries 16 and 18 converted into a 100% share of the portfolio of enterprises.

2 The HHI as measured by the number of loans (HHI numbers) is defined as the sum of the squared relative shares of the numbers of loans of individual industries in the total number of loans.

CHART 5
CONTRIBUTIONS OF THE MAIN SECTORS TO THE HHI ACROSS BANK GROUPS

(index)



Source: CNB, authors' calculations

also been gaining in significance, partly due to the funding of photovoltaic power station construction. The risks in this area include long investment horizons, innovative technology and legislative and tax changes. Supervisors should therefore systematically evaluate the credit quality of this portfolio.

According to our HHI analysis of sector concentration, the aggregate values for the benchmark portfolio indicate relatively low concentration. One significant systemic structural shift is a change in the position of "Real estate activities", the industry with the largest exposure accompanied by a high level of concentration, especially in the small bank segment. In the medium-sized bank segment, sector concentration is rising significantly in the direction of non-residents ("Unassigned"). The dynamics and internal structure of this concentration therefore

warrant increased attention.

5. USE OF THE FACTOR MODEL TO SET CAPITAL REQUIREMENTS FOR SECTOR CONCENTRATION RISK³

In this section, we complement the simple non-model-based approach to determining the level of undiversified idiosyncratic risk with a model-based approach that allows us to take sector concentration risk into account when setting economic capital.

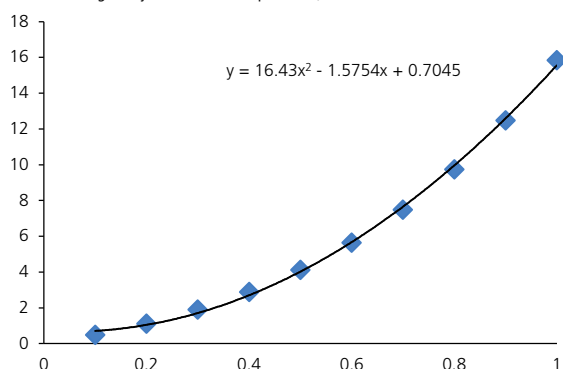
Sector concentration is a typical risk component of loans to non-financial corporations. The risk stems from dependencies arising between firms due to their sector affiliation and the prevailing economic environment.

The asymptomatic single risk factor (ASRF) model, which forms the basis for determining risk weights in the IRB approach, assumes that all loans are dependent on the same systemic risk factor. This ensures that economic capital can be determined individually for each and every loan without taking the portfolio structure into account. Given the assumed single correlation structure, the credit risk of a portfolio with a different sector structure may be either overestimated or underestimated. The risk contribution of sector concentration to the total portfolio risk can only be accounted for by extending the modelling framework.

³ Our calculations are based on the model presented in Düllmann and Masschelein (2007).

CHART 6
SENSITIVITY OF ECONOMIC CAPITAL TO CHANGE IN THE SECTOR FACTOR WEIGHT

(x-axis: factor weight rs; y-axis: economic capital in %)



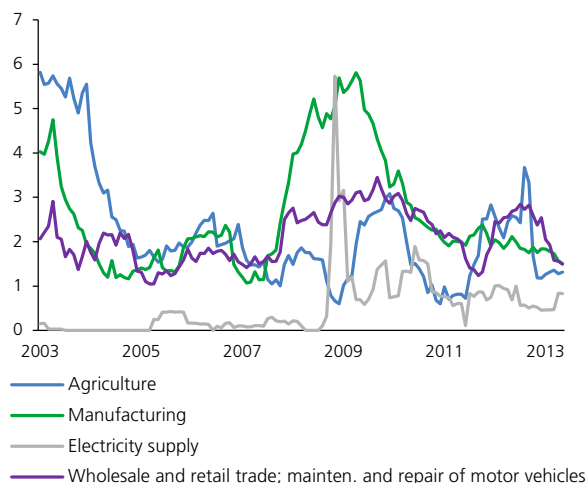
Source: Authors' calculations

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CHART 7

DEFAULT RATES BY SECTOR IN 2003–2013 (MAXIMUM < 6%)

(12-month default rate in %)

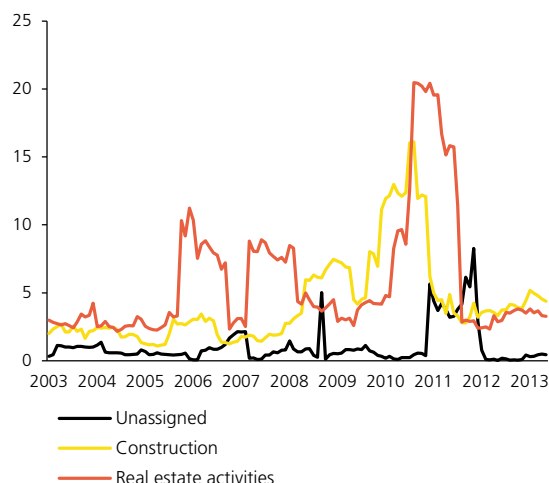


Source: CNB

CHART 8

DEFAULT RATES BY SECTOR IN 2003–2013 (MAXIMUM > 6%)

(12-month default rate in %)



Source: CNB

A key prerequisite for measuring sector concentration is the industry classification. The CCR data contain NACE industry code information for residents, and the breakdown into economic activities is based on that information. The industry definition allows individual risk factors to be assigned directly. The ideal industry classification is one where the intra-sectoral correlation of asset yields is high and the inter-sectoral correlation is low. However, this breakdown cannot be used in practice because the intra and inter-sectoral correlations are both heterogeneous. The intra-sectoral asset correlation is usually described by statistically calibrated functions such as non-financial corporation turnover, while the inter-sectoral correlation tends to be estimated, for example, by the time series of correlations of relevant stock indices.

As we do not have all the necessary information available, we use an intra-sectoral correlation of 0.25 as implied by the sector factor weight of 0.5 found in foreign empirical studies (Hahnenstein, 2004, and Lopez, 2004) and we replace the correlation of the relevant stock indices by the correlation of the year-on-year change in the sector's quarterly contribution to GDP growth. To assess the impact of this factor on the calculation results, we tested the sensitivity of economic capital to change in the sector factor weight. The results are shown in Chart 6, in which economic capital increases exponentially with increasing factor weight. From this perspective, we consider the value of 0.5 to be reasonable.

Another component of the model is the loan default rate for the relevant sector, which was calculated using CCR

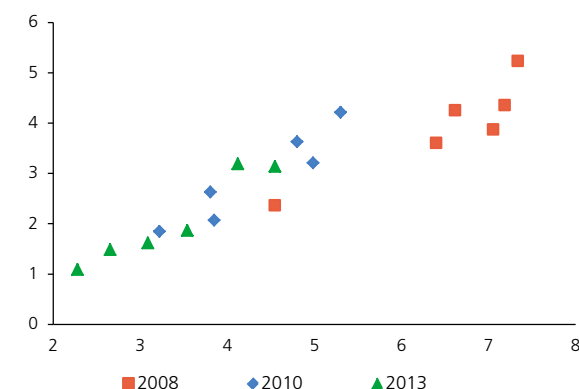
data (see the thematic article *Use of the Czech Central Credit Register for financial stability purposes* in this Report). Charts 7 and 8 plot the default rates over the period of 2004–2013. The charts clearly show the effect and aftermath of the economic crisis in 2008–2009, when the highest default rates were recorded in “Manufacturing”, “Wholesale and retail trade” and “Electricity supply”, and in 2010, when “Construction” and “Real estate activities” attained the highest default rates. The crisis can be seen fading away in 2011 and 2012, except in “Wholesale and retail trade”, where we see renewed growth in the default rate. During the crisis, “Construction”, “Manufacturing” and “Real estate activities” all recorded their highest ever absolute default rates.

The output of the model is the need for economic capital, which proxies for regulatory capital in the model context. The results show that the economic capital for credit risk inclusive of sector concentration risk of the benchmark portfolio (the aggregate CCR) and the model portfolios differs for different degrees of portfolio diversification depending on the level of risk of each sector (see Chart 9). The need for economic capital increases with increasing concentration in sectors with higher default rates and with general growth in the default rate.

The relationship between concentration as measured by the HHI and economic capital (see Chart 10) reveals that the difference in the capital requirements of the least and most concentrated portfolios in periods when the portfolio default rate is low (2010 and 2013) is roughly 2 percentage

CHART 9

WEIGHTED PORTFOLIO DEFAULT RATES VERSUS ECONOMIC CAPITAL FOR THE BENCHMARK AND MODEL PORTFOLIOS
(x-axis: economic capital in %; y-axis: weighted portfolio default rate in %)



Source: CNB, authors' calculations

points, signalling a need for additional economic capital to cover sector concentration risk.

Although we observe some dependence between the level of concentration as measured by the HHI and economic capital, the results simultaneously show that the HHI cannot be used to proxy for the capital requirement, as it does not respond to the default rate in individual sectors. This can be seen in Chart 10 on the example of the low risk of a more concentrated portfolio (2008 top left) and the higher risk of a less concentrated portfolio (bottom right). However, the HHI may be a suitable auxiliary indicator of economic capital in periods when the default rate is generally low and not too different.

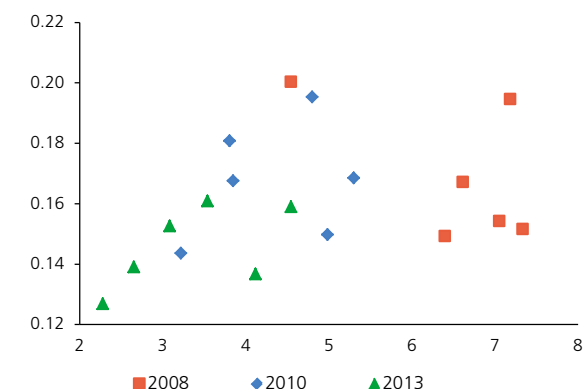
6. CONCLUSION

Concentration of credit risk in asset portfolios is one of the potential causes of large losses in credit institutions and one of the specific risks subject to supervisory review under Pillar 2. Large exposure risk is addressed in quite some detail in the CRR, but sector concentration is not subject to detailed regulation.

This article therefore presented a method for evaluating whether sector concentration, measured at the level of the main sectors of the national economy, indicates an additional need for capital under Pillar 2. We then measured the degree of concentration and its long-term tendencies using the HHI as a recommended simple non-model-based approach. We worked with information on the portfolio of loans to resident and non-resident legal entities taken from the Czech Central Credit Register (CCR), which we

CHART 10

HHI VERSUS ECONOMIC CAPITAL FOR THE BENCHMARK AND MODEL PORTFOLIOS
(x-axis: economic capital in %; y-axis: HHI)



Source: CNB, authors' calculations

incorporated into a benchmark portfolio (the aggregate CCR exposure) and model portfolios (simulating characteristics of portfolios in the banking sector).

The factor model used for the indicative calculation of economic capital signals a need for higher economic capital for model portfolios with higher sector concentrations and riskier sector profiles (higher loss given default). The results of the model for 2013 indicate that the difference in the capital requirements of the least and most concentrated portfolios as measured by the HHI is 2 percentage points. In such cases, supervisors should evaluate whether sector concentration risk is correctly measured, assessed and incorporated into capital requirements in the bank's internal capital adequacy assessment process (ICAAP). The results of the model can be used as a guide for assessing the positions of individual banks and setting a consistent approach to defining the amount of capital required for this risk at the level of the banking sector as a whole.

The aggregate HHI value for the concentration of the benchmark portfolio (0.135–0.155) can be regarded as quite low and comparable with that for Germany. Its internal structure has seen a significant change in the form of long-term growth in the exposure of the "Real estate activities" sector, which has become dominant in the present decade. This change needs to be reflected both in supervisory practice (the assessment of capital requirements under Pillar 2) and in stress testing and the design of macroprudential tools. Growth in the exposure of borrowers unassigned to a sector (primarily non-residents) and the significant share of such borrowers in the segment of medium-sized banks also warrant attention.

We also assess the possibility of using the HHI for setting capital requirements. The HHI cannot be used directly because it does not take account of sector risk. However, it can be used as an auxiliary indicator, as under certain conditions a higher HHI indicates higher capital requirements.

The method described in this article for indicating additional capital requirements for sector concentration risk and measuring the level and evolution of concentrations based on the HHI can be used as an auxiliary tool for supervisory purposes and in the design of macroprudential measures until the unified methodology being prepared by the EBA's Advanced Working Group on Sectoral Risk becomes available. This methodology is planned to be submitted for discussion in April 2015.

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